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WildfireIQ Kamloops: An Open Platform for Wildfire Risk, Air Quality, Climate Trends, and Community Preparedness in British Columbia, with a Data-Grounded Assistant

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Abstract

Background and objectives. Wildfire seasons in British Columbia have grown in scale and consequence, and the Thompson-Okanagan region around Kamloops has been among the most affected in Canada over the past decade. The 2023 season was the largest on record nationally. Smoke exposure is now a recurring public-health event, yet the information available to residents remains scattered: air-quality apps do not connect to fire activity, evacuation notices are static, and no public tool ties fire risk, smoke, long-term climate change and household preparedness together for this region. WildfireIQ Kamloops set out to close that gap with one open, free-to-run web platform built entirely from public data, and to do so with transparent, held-out validation of every prediction it shows.

Methods. The platform is built as a single backend process (Python, FastAPI) and a browser front end (React, TypeScript, CesiumJS). Seventeen scheduled jobs pull from the BC Wildfire Service, BC Emergency Management and Climate Readiness, NASA FIRMS, Environment and Climate Change Canada, Open-Meteo and WAQI on cadences from five minutes to nightly, writing one Parquet file per dataset; the API serves from those files, so the platform keeps working when a source is down. Two models were trained. The wildfire-risk model is a single LightGBM classifier pooled across four regions (Thompson-Okanagan, Central Okanagan, Lower Mainland, Prince George), predicting the probability that at least one fire ignites in a region on a given day from that region's ERA5 weather, Canadian Fire Weather Index codes computed with the Van Wagner equations, lagged and rolling weather, drought signals and a base-rate prior; the regional probability is scaled by each H3 hexagon's historical fire density into a four-class map. It was trained on 1999–2021, calibrated on 2022, and tested on a fully unseen 2023, with results reported per region. The air-quality model is 21 LightGBM quantile regressors (seven horizons to 48 hours, three quantiles) over hourly CAMS PM2.5 and co-located weather at Kamloops. A fifth module was added after the four planned features were complete: an assistant that answers only through 25 read-only tools over the platform's own data, primed with a live situation brief, using GLM 5.3 Flash via OpenRouter behind rate and spend limits and a deterministic emergency-wording tripwire.

Results. All five modules are built, connected to live data, and covered by 158 backend and 36 frontend tests. On the unseen 2023 season the risk model reaches a precision-recall AUC of 0.72 for the Thompson-Okanagan, 0.61 for Prince George and 0.51 for the Central Okanagan against a Fire-Weather-Index baseline of 0.37 and a climatology baseline of 0.19; the Lower Mainland, a low-event area with 35 fire-days in the test year, scores

0.29 and is disclosed as such. Pooled ROC-AUC is 0.874 and the calibrated Brier score 0.108. The air-quality forecaster beats a persistence baseline at five of seven horizons. Over 1999–2025 the Thompson-Okanagan record shows a July mean daily-maximum temperature trend of +0.18 °C per year (95% CI 0.06–0.32), rising vapour-pressure deficit, and +1.95 high-fire-danger days per year (95% CI 1.0–2.75), all distinguishable from noise by Theil-Sen estimation with bootstrap intervals; summer precipitation, area burned and season length are not. The assistant passes a 32-case live evaluation spanning every data surface and a set of misuse attempts, at a measured \$0.0013 and a median of six seconds per answer.

Conclusions. WildfireIQ shows that one small team, using only free and open data, can deliver a validated, honest wildfire information platform for a region, and that a language-model assistant can be made useful and safe by grounding it exclusively in the platform's own tools rather than its training data. Regions are configuration rather than code: adding a city to the risk model is a data step, and the components tied to British Columbia are enumerated so the platform can be carried to other provinces. Known limits are stated in the software itself: one weather series per region, a single-point air-quality forecast, a placeholder in place of the live climate-model ensemble, and a model that is informational and never a substitute for the BC Wildfire Service or emergency officials. The code, models, model cards and documentation are released under the MIT licence.

Keywords: wildfire risk prediction; air quality forecasting; Fire Weather Index; LightGBM; quantile regression; H3 hexagonal grid; community preparedness; FireSmart; climate trends; Theil-Sen; LLM agent; tool use; British Columbia; Thompson-Okanagan; environmental informatics; open data

The platform at a glance

Item	Figure
Modules	Globe · Air Quality · Preparedness · Climate Trends · Assistant
Public data sources	BC Wildfire Service, BC EMCR, NASA FIRMS, ECCC GeoMet, Open-Meteo, WAQI, FireSmart Canada, Health Canada
Scheduled data jobs	17 (15 recurring, from every 5 minutes to nightly; 2 one-time)
Historical fire records	96,356 incidents, province-wide, 1999 to today
Weather history	27 years of ERA5 reanalysis per region, extended daily
Risk model	Pooled LightGBM, 4 regions, 523 H3 hexagons, 42 features
Held-out 2023 PR-AUC by region	0.72 · 0.61 · 0.51 · 0.29 (FWI baseline 0.37)
Air-quality model	21 LightGBM quantile models, 7 horizons to 48 h
Assistant	25 tools, GLM 5.3 Flash via OpenRouter, 32/32 live evaluation, \$0.0013 per answer
Automated tests	158 backend, 36 frontend

Source code, model cards and the full documentation set: github.com/DeeparshSingh/WildfireIQ. Every figure above is taken from the released build of 8 September 2026.